



RAJEEV GANDHI MEMORIAL COLLEGE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING
PROJECT ABSTRACT

BATCH NO : 14

TITLE: “A Robust Chaotic Encryption Scheme for Facial Images Using Deep Face, PSO, and STP Theory”

Abstract:

In this project, a novel image encryption scheme is proposed that integrates chaotic systems, Deep Face-based facial recognition, Particle Swarm Optimization (PSO), and Semi-Tensor Product (STP) theory to achieve efficient and secure selective facial image encryption. A new 3D Coupled Ikeda Map with Bounded Amplitude (3D-CIMBA) is introduced, exhibiting strong chaotic properties and controllable Lyapunov exponents. PSO is used to optimize the chaotic system's control parameters based on the input image, enhancing key sensitivity and resistance to known-plaintext attacks. The Deep Face model ensures accurate localization of facial regions, which are then encrypted using a combination of row-column cyclic shifting and STP-based diffusion. Cryptographic analyses, including entropy, correlation, histogram, NPCR, UACI, and robustness against noise and cropping, confirm the high security, low computational overhead, and fast performance of the scheme. Experimental results will show that the proposed method outperforms traditional image encryption algorithms, especially in selective encryption scenarios.

Keywords: Chaotic image encryption, 3D-CIMBA, Deep Face, particle swarm optimization, semi-tensor product theory, selective encryption, facial region recognition, key optimization, Lyapunov exponent, cryptographic security.

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